Do Now · retrieval
Programming paradigms

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is an object?

Q2

What does a constructor do?

Q3

What does a return statement do?

Q4

What property distinguishes a pure function from a method that changes a shared counter?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Procedural — Instructions and procedures operate on arguments or shared data. A function can be reused independently.

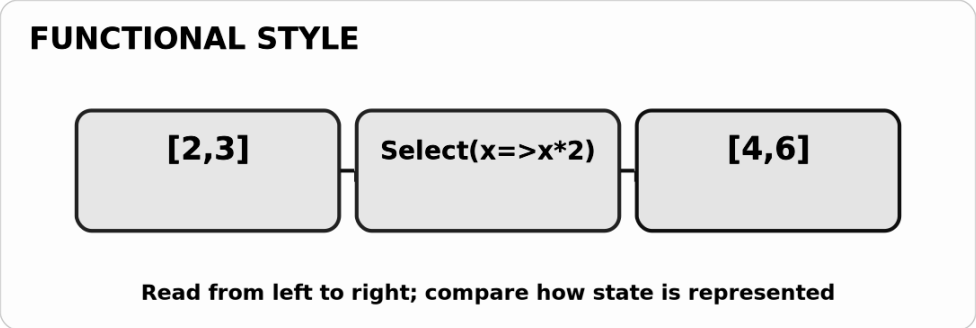
Core idea 2

Object-oriented — The Account object combines state (Balance) and behaviour (Deposit). With inheritance, a subclass can reuse and specialise a class. A virtual method overridden in a derived class supports polymorphism: the actual object decides which implementation runs.

Core idea 3

Functional style — Values are transformed using expressions/functions, often emphasising avoiding mutation of existing input data.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

paradigm — Overall style for organising programs.

procedural — Organises a program around sequences of operations and subroutines.

object-oriented — Encapsulates state and behaviour into objects.

functional — Emphasises functions applied to values and reduced side effects.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Classify a function Square(x) that returns x*x without storing object state.

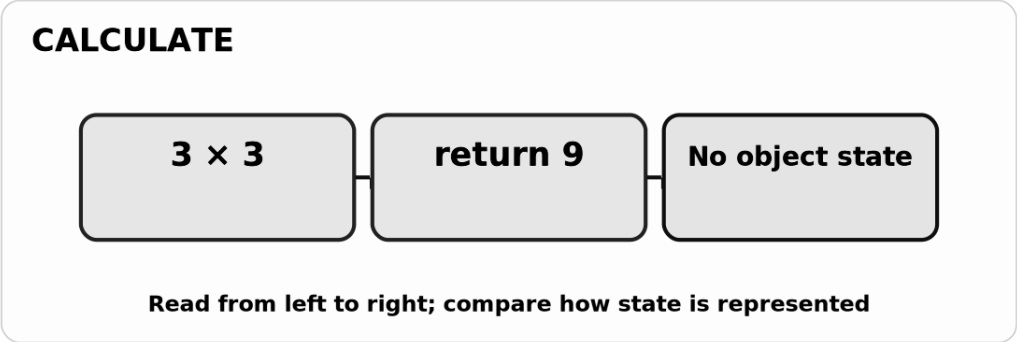
Follow the worked method

1. Predict — It is procedural when used as a standalone function in a procedural programme.

2. Follow the declared and actual types — Both variables are declared as BaseMessage . Their actual objects are EmailMessage and TextMessage , each inheriting from the base class.

3. Output and design consequence — Outputs Email then Text . The caller can add another derived message class without writing separate type-checking branches.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

The Score object holds Points alongside a method that changes its state.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Classify a program using standalone functions with arguments and returns. Give one advantage.

Question 2

Compare procedural and object-oriented account implementations, then explain how an overridden method on a derived SavingsAccount can be called through an Account reference.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why `values.Select(x=>x*2)` can be considered functional-style programming, and one limitation of labelling C# purely functional.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A language such as C# can support both procedural and object-oriented programming.

Use key vocabulary from memory

1. Explain paradigm in your own words.

2. Explain procedural in your own words.

3. Explain object-oriented in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.